Chemistry - Undergraduate Level Assessment 24 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is lower for argon than for neon?
   - (a) Melting point
   - (b) Boiling point
   - (c) Polarizability
   - (d) First ionization energy

2. When the following equation is balanced, which of the following is true? ___ MnO 4- + ___ I- + ___ H+ <-> ___ Mn_2+ + ___ IO 3- + ___ H_2O
   - (a) The I- : IO 3- ratio is 3:1.
   - (b) The MnO 4- : I- ratio is 6:5.
   - (c) The MnO 4- : Mn_2+ ratio is 3:1.
   - (d) The H+ : I- ratio is 2:1.

3. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
   - (a) 23.56 GHz
   - (b) 42.58 MHz
   - (c) 74.34 kHz
   - (d) 13.93 MHz

4. Of the following compounds, which has the lowest melting point?
   - (a) HCl
   - (b) AgCl
   - (c) CaCl 2
   - (d) CCl 4

5. Of the following compounds, which is LEAST likely to behave as a Lewis acid?
   - (a) BeCl 2
   - (b) MgCl 2
   - (c) ZnCl 2
   - (d) SCl 2

True / False (10 marks)

Answer True or False

6. True or False: Most organic compounds do not contain silicon atoms, making tetramethylsilane a suitable reference for 1 H NMR chemical shifts.

7. True or False: The hybrid sp^3 orbitals for carbon are constructed from a linear combination of four atomic orbitals.

8. True or False: The 13 C NMR spectrum of hexane exhibits five distinct chemical shifts because of its symmetrical structure.

9. True or False: Protons in a molecule with a rotational correlation time of 1 ns have the fastest spin-lattice relaxation at a magnetic field of 3.74 T.

10. True or False: 0.100 M NaCl has a higher ionic strength than 0.050 M AlCl 3 due to its higher concentration of ions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the electron affinity of atoms, and analyze why calcium has the lowest electron affinity among its peers.

12. Discuss the concept of conjugate acid-base pairs in chemistry, using H_3O+ and H_2O as examples. Explain their roles and how they interact in acid-base reactions.

End of Paper